

ROUNDWORMS

The simple, cylindrical roundworms are remarkably successful – surviving almost everywhere, resisting drought, and reproducing rapidly.

The phylum Nematoda is ubiquitous. There can be millions of roundworms in a square metre of soil, and they also live in freshwater or marine habitats.

Many are parasites. These worms, or nematodes, can be extraordinarily prolific, producing hundreds of thousands of eggs per day. When environmental conditions deteriorate they can survive heat, frost, or drought by producing resistant stages and becoming dormant.

Roundworms have a muscle-lined body cavity and two openings to their gut – the mouth and anus. Their cylindrical body is coated in a tough layer, called a cuticle – functionally similar to that of arthropods – which they moult periodically as they grow.

PHYLUM	NEMATODA
CLASSES	2
ORDERS	17
FAMILIES	About 160
SPECIES	About 26,000



BLACK AND YELLOW FLATWORM

Pseudoceros dimidiatus
F: Pseudocerotidae

Marine polyclads are large free-living flatworms. Many are strikingly coloured, warning predators of their nasty taste. This species is from Indo-Pacific waters.

7–8 cm
2¾–3¼ in



SHOVEL-HEADED LAND FLATWORM

Bipalium kewense
F: Bipaliidae

Land flatworms are mostly tropical and need damp conditions. This Asian species has been accidentally introduced to greenhouses around the world.

20–30 cm
8–12 in



FLOWERS' FLATWORM

Pseudobiceros flowersi
F: Pseudocerotidae

Like many reef-dwelling polyclads, this Indo-Pacific species glides with an undulating margin. It has been found under rubble in lagoons.

4–9 cm
1½–3½ in



GOLD-SPECKLED FLATWORM

Thysanozoon nigropapillosum
F: Pseudocerotidae

Many *Thysanozoon* polyclads are coated with pimply swellings, which are yellow-tipped on this otherwise velvety black species of the Indo-Pacific.

7–8 cm
2¾–3¼ in

1.5–2 cm
½–¾ in



DRAB FRESHWATER FLATWORM

Dugesia lugubris
F: Planariidae

Triclad flatworms have a triple-branched gut. This European species is one of many triclad flatworms found in freshwater habitats; others are marine.

1–1.5 cm
⅜–½ in



BROWN FRESHWATER FLATWORM

Dugesia tigrina
F: Planariidae

This species is native to freshwater habitats of N. America, but has been introduced to Europe.

NEW ZEALAND LAND FLATWORM

Arthurdendyus triangulatus
F: Geoplanidae

This large soil-living species is native to New Zealand, but has invaded Europe. It preys on earthworms.

10–17 cm
4–6½ in



STREAM FLATWORM

Dugesia gonocephala
F: Planariidae

Many of the freshwater *Dugesia* flatworms, such as this European species of running water, have ear-like flaps for detecting water currents.

2–3 cm
¾–1¼ in

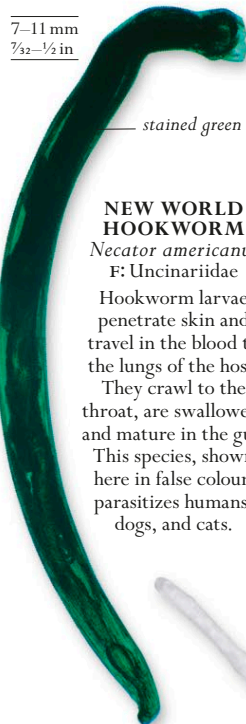


NEW WORLD HOOKWORM

Necator americanus
F: Uncinariidae

Hookworm larvae penetrate skin and travel in the blood to the lungs of the host. They crawl to the throat, are swallowed, and mature in the gut. This species, shown here in false colour, parasitizes humans, dogs, and cats.

7–11 mm
⅞–½ in

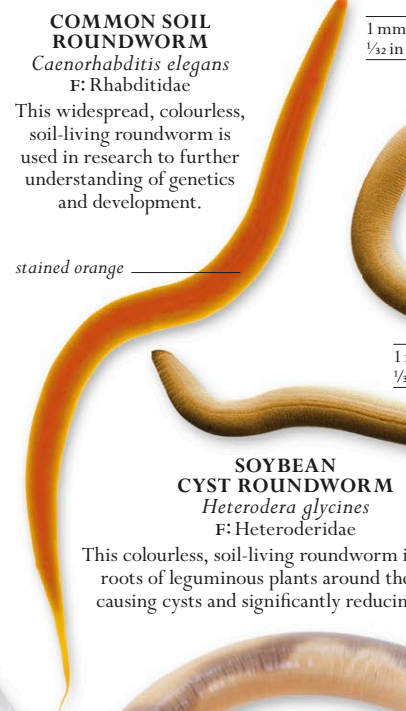


COMMON SOIL ROUNDWORM

Caenorhabditis elegans
F: Rhabditidae

This widespread, colourless, soil-living roundworm is used in research to further understanding of genetics and development.

1 mm
⅓ in



SOYBEAN CYST ROUNDWORM

Heterodera glycines
F: Heteroderidae

This colourless, soil-living roundworm infects the roots of leguminous plants around the world, causing cysts and significantly reducing yields.

1 mm
⅓ in



3–5 cm
1¼–2 in

COMMON WHIPWORM

Trichuris trichiura
F: Trichuridae

Like many other gut parasites, the mainly tropical whipworm infects humans who eat food contaminated with faeces. It completes its life cycle in the intestines.



LARGE INTESTINAL ROUNDWORM

Ascaris lumbricoides
F: Ascarididae

A common parasite of humans in regions of poor sanitation where people get infected by eating food contaminated with faeces, with adult worms then living in the host intestine.

15–35 cm
6–14 in